CLINICAL · NOSE & SINUSES

Hereditary haemorrhagic telangiectasia

Hereditary haemorrhagic telangiectasia is a genetic disease of blood vessels that presents to the rhinologist as nosebleeds that never stop coming back. The nose is the visible part of a systemic disorder: every patient also needs screening for silent arteriovenous malformations in the lungs and brain.

AT A GLANCE

- Recurrent bleeds, telangiectasia and family history.
- Screen for lung and brain AVMs.
- Use gentle care; avoid repeated cautery.
- Three genes (ENG, ACVRL1, SMAD4); step up from KTP laser to septodermoplasty to nasal closure; resorbable packs only.

1 CHAPTER ONE Overview

A vascular disease that bleeds through the nose.

1.1 Definition

HHT, also called *Osler-Weber-Rendu syndrome*, is an autosomal dominant disorder of angiogenesis. It produces fragile *telangiectasias* in the skin and mucous membranes and larger *arteriovenous malformations* (AVMs) in internal organs.

1.2 Epidemiology

About 1 in 5,000 to 10,000 people, reported mainly in white populations, affecting men and women equally. Penetrance increases with age.

1.3 Why it matters

Recurrent epistaxis causes chronic iron-deficiency anaemia and repeated transfusions, and treating it badly (repeated packing and aggressive cautery)

makes it worse. Pulmonary AVMs cause paradoxical embolism, stroke and brain abscess; cerebral AVMs can bleed.

2 CHAPTER TWO Pathogenesis

A defect in vessel signalling.

2.1 Genetics

Mutations disturb transforming growth factor-beta signalling in endothelial cells, so the connective tissue and matrix that should support new vessels during angiogenesis fail to form.

- **HHT type 1:** mutation in *endoglin* (chromosome 9q34); pulmonary and cerebral AVMs are commoner.
- **HHT type 2:** mutation in *ACVRL1*, also called ALK1 (chromosome 12q); hepatic AVMs are commoner.
- **Juvenile polyposis–HHT overlap:** mutation in *SMAD4*; these patients need colonoscopy because of colonic polyps and cancer risk. Together, the three genes explain about 90% of clinically diagnosed cases, and genetic testing is mainly used to screen relatives.

2.2 The lesion

Postcapillary venules dilate and connect directly to dilated arterioles, losing the capillary bed between them. The walls lack elastic tissue and smooth muscle, so they cannot contract after injury: minor trauma from dry air or a sneeze causes brisk bleeding that does not stop easily.

3 CHAPTER THREE Clinical features

Nose, skin and organs.

3.1 Epistaxis

The commonest and earliest feature: recurrent, spontaneous bleeding from childhood, from both sides, becoming more frequent with age. About 80 to 90% have nosebleeds by age 21.

3.2 Mucocutaneous telangiectasias

Small red spots that blanch with pressure, on the lips, tongue, palate, fingertips, face and nasal mucosa. About 80 to 90% develop them by their thirties or forties.

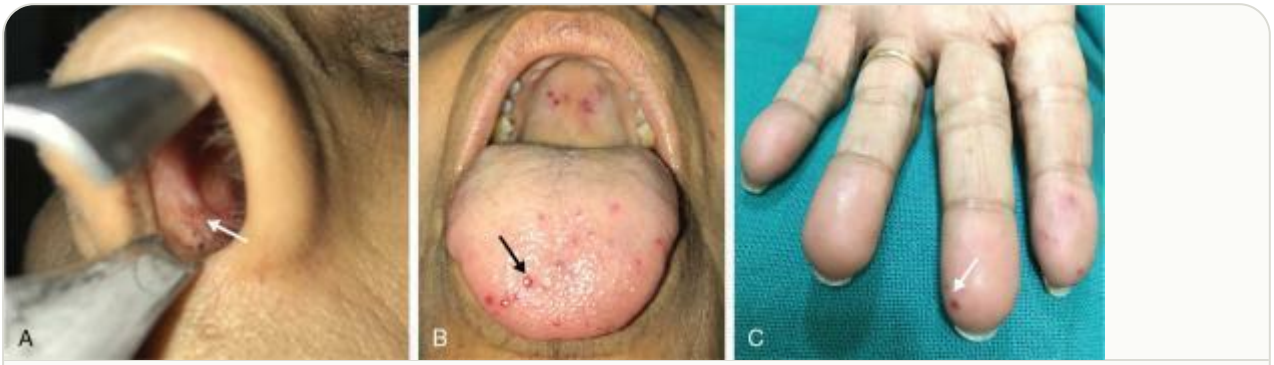


FIGURE 1 A patient with recurrent epistaxis from HHT, with telangiectasias on the nasal septum, tongue, palate and fingertips (arrows).
Dhingra, fig. 33.3

3.3 Visceral arteriovenous malformations

ORGAN INVOLVEMENT IN HHT (LEE)

ORGAN	FREQUENCY	WHAT IT CAUSES
Lungs	About 30%	Hypoxia, paradoxical emboli, stroke, brain abscess, haemoptysis
Brain	5–15%	Haemorrhage, seizures
Liver	40–70%	Usually silent; high-output heart failure, portal hypertension

ORGAN	FREQUENCY	WHAT IT CAUSES
Gut	20–25%	Bleeding and anaemia
Spine	Rare	Spinal haemorrhage

3.4 Anaemia

Chronic iron-deficiency anaemia from nasal and gut bleeding.

4 CHAPTER FOUR Diagnosis

The Curaçao criteria.

4.1 Criteria

- **Epistaxis:** spontaneous and recurrent.
- **Telangiectasias:** multiple, at characteristic sites (lips, mouth, fingers, nose).
- **Visceral lesions:** gastrointestinal telangiectasia, or pulmonary, hepatic, cerebral or spinal AVMs.
- **Family history:** a first-degree relative with HHT.

4.2 Interpretation

Three or more criteria: *definite*. Two: *possible* or suspected. Fewer than two: unlikely. Genetic testing confirms the diagnosis in uncertain cases and in relatives.

4.3 Work-up

- Full head and neck examination with nasal endoscopy.
- Full blood count and iron studies.
- **Contrast echocardiography** (bubble study) to screen for pulmonary AVMs, in every patient.
- **MRI brain with contrast** for cerebral AVMs.

- Endoscopy for symptomatic gastrointestinal bleeding; Doppler ultrasound or CT for the liver.
- Genetic counselling.

SCREEN THE LUNGS

A pulmonary AVM can cause a stroke or brain abscess in a patient who has only ever complained about nosebleeds. Screen every patient with contrast echocardiography and treat significant AVMs by embolisation.

4.4 Differential diagnosis

Bleeding disorders (von Willebrand disease, platelet disorders), anticoagulant use, and a bleeding nasal tumour in one-sided bleeding.

5 CHAPTER FIVE Management of the nose

Reduce bleeding without damaging the mucosa.

5.1 Principles

The goal is fewer and smaller bleeds and a better haemoglobin, not cure. Avoid trauma to the mucosa: repeated packing and aggressive cautery destroy healthy mucosa, encourage new telangiectasias and can cause septal perforation.

5.2 The acute bleed

Gentle pressure; resorbable packing if needed, because removing a conventional pack tears lesions and restarts bleeding; avoid cautery of large areas.

5.3 Conservative and medical treatment

1. **Humidification and emollients**
   Saline sprays, ointments and humidified air keep the mucosa moist.

2 **Iron and transfusion**
Correct anaemia; intravenous iron when oral iron fails.

3 **Tranexamic acid**
Oral antifibrinolytic reduces the frequency and severity of bleeds.

4 **Hormonal treatment**
Oestrogen or tamoxifen convert the mucosa to a thicker squamous type.

5 **Bevacizumab**
An anti-VEGF antibody, given topically, by submucosal injection or intravenously, reduces bleeding in severe disease.

5.4 Procedures

5.4.1 Photocoagulation

KTP, argon, Nd:YAG or diode laser, bipolar diathermy or coblation, aimed at individual lesions and repeated every few months as new ones appear.

5.4.2 Septodermoplasty

The bleeding anterior septal mucosa is replaced with a split- or full-thickness skin graft, which is tougher. It halves the number of later laser treatments.

5.4.3 Young's procedure

Closure of the nostrils, one or both sides, by suturing flaps inside the vestibule. Bleeding stops in about 83%, at the price of nasal breathing and smell. Rebleeding after reopening is troublesome.

5.4.4 Embolisation

Gives only temporary benefit because new collaterals form; reserved for emergencies.

OPERATIVE GUIDES IN THEATRE 1 operation

Surgical Management of Hereditary Hemorrhagic Telangiectasia
Laser, septodermoplasty, Young's closure

PEARLS

- Repeated cautery makes HHT worse.
- Resorbable packs, never conventional ribbon gauze, if packing is unavoidable.

- Treat the nose and the haemoglobin together.

A STAGED APPROACH (OXFORD TEXTBOOK)

Treatment is stepped up by severity, judged by bleeding frequency, the haemoglobin and transfusion needs; the Epistaxis Severity Score helps track it. The first intervention is usually an examination under anaesthesia with **KTP or argon laser** to stage and treat the disease: these green wavelengths are absorbed by haemoglobin, each lesion is circled and then ablated, tranexamic acid and antibiotic cover are given at induction, and no pack is needed. Bipolar diathermy works better than laser in a bloody field. If laser controls bleeding for less than about three months, move to **septodermoplasty** or systemic treatment; benefit lasts months after laser, years after septodermoplasty, and permanently after **nasal closure** (the Lund-modified Young's procedure with three vestibular flaps), which is offered to transfusion-dependent patients who accept the loss of nasal breathing and smell.

In an acute bleed, bleeding is almost always from the anterior third, so posterior packs are rarely needed. Use resorbable packs such as Nasopore or Surgicel, or a haemostatic matrix, rather than expanding sponges that tear other lesions; give antibiotic cover for any pack; and if packing fails, embolise the facial and maxillary feeders without coils. Recent trials of systemic anti-angiogenic drugs, including intravenous bevacizumab and oral pomalidomide, have reduced bleeding in severe disease, and these are now used by specialist HHT centres.

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Inheritance and genes:** autosomal dominant; endoglin (type 1) and ACVRL1 or ALK1 (type 2).
- **Curaçao criteria** and how many make the diagnosis definite.
- **Screening** for pulmonary and cerebral AVMs.

- **The treatment ladder,** septodermoplasty and Young's procedure.

6.2 Common pitfalls

COMMON PITFALLS

- Treating every bleed as simple epistaxis without asking about family history and looking at the lips and fingers.
- Repeated packing and wide cautery.
- Forgetting to screen the lungs.

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